the practical device that came to bear his name. By leaning heavily on Joseph Henry's understanding of the laws of electricity and magnetism, Samuel Morse devised his profitable version of the telegraph.

Inventors and physicists alike took pride in the perfection of their understanding. Everything in the heavens and on Earth seemed to obey the Newtonian laws of physics, and mastery of the laws was bringing humans a mastery of their environment—and perhaps one day would bring mastery of the entire Universe.

All the old, well-established Newtonian laws and their technological applications Einstein could learn in Heinrich Weber's lectures, and learn well. Indeed, in his first several years at the ETH, Einstein was enthusiastic about Weber. To the sole woman in his ETH class, Mileva Maric (of whom he was enamored), he wrote in February 1898, "Weber lectured masterfully. I eagerly anticipate his every class."

But in his fourth year at the ETH Einstein became highly dissatisfied. Weber lectured only on the *old* physics. He completely ignored some of the most important developments of recent decades, including James Clerk Maxwell's discovery of a new set of elegant electromagnetic laws from which one could deduce *all* electromagnetic phenomena: the behaviors of magnets, electric sparks, electric circuits, radio waves, light. Einstein had to teach himself Maxwell's unifying laws of electromagnetism by reading up-to-date books written by physicists at other universities, and he presumably did not hesitate to inform Weber of his dissatisfaction. His relations with Weber deteriorated.

In retrospect it is clear that of all things Weber ignored in his lectures, the most important was the mounting evidence of cracks in the foundation of Newtonian physics, a foundation whose bricks and mortar were Newton's concepts of space and time as absolute.

Newton's *absolute space* was the space of everyday experience, with its three dimensions: east—west, north—south, up—down. It was obvious from everyday experience that there is one and only one such space. It is a space shared by all humanity, by the Sun, by all the planets and the stars. We all move through this space in our own ways and at our own speeds, and regardless of our motion, we experience the space in the same way. This space gives us our sense of length and breadth and height; and according to Newton, we all, regardless of our motion, will agree on the length, breadth,